

function of two variables.

A function of two variables x & y is a rule that assigns a unique real number $f(x, y)$ to each point (x, y) in some set D in xy -plane.

$$Q \ f(x, y) = x^2y + 1$$

$$(a) \ f(2, 1) = 5$$

$$(b) \ f(1, 2) = 3$$

$$(c) \ f(0, 0) = 1$$

$$(d) \ f(1, -2) = -1$$

$$(e) \ f(4a, a) = 16a^3 + 1$$

$$(f) \ f(ab, a-b) =$$

$$= (a^2b^2)(a-b) + 1$$

$$= a^3b^2 - a^2b^3 + 1$$

$\approx$

$$\textcircled{1} f(x) = (2)^2(1) + 1$$

$$= 4 + 1 = 5$$

$$\textcircled{2} f(x) = (1)^2(2) + 1$$

$$= 2 + 1 = 3$$

$$\textcircled{3} f(x) = (0)^2(0) + 1$$

$$= 1$$

$$\textcircled{4} f(x) = (1)^2(-2) + 1$$

$$= -2 + 1 = -1$$

$$\textcircled{5} f(x) = (4a)^2(a) + 1$$

$$= 4016a^3 + 1$$



$$f(xy) = x + \sqrt[3]{xy}$$

$$(a) f(t^4t^5)$$

$$= t + [t(t^5)]^{1/3}$$

$$= t + (t^6)^{1/3}$$

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$$\frac{(8y^3)^{1/3}}{(2^3)^{1/3}}$$

$$= \frac{t(t+1)}{2}$$

$$(b) f(x, x^5)$$

$$= \frac{x + \sqrt[3]{x \cdot (x^5)}}{3}$$

$$= \frac{x + (x^6)^{1/3}}{3}$$

$$= \frac{x + x^2}{3}$$

$$= \frac{x(x+1)}{3}$$

$$(c) f(2y^2, 4y)$$

$$= \frac{2y^2 + [(2y^2)(4y)]^{1/3}}{3}$$

$$= \frac{2y^2 + (8y^3)^{1/3}}{3}$$

$$= \frac{2y^2 + 2y}{3}$$

$$= \frac{2y(y+1)}{3}$$

$$g f(x, y) = xy + 3$$

$$(a) f(x+y), (x-y)$$

$$(b) f(xy, 3x^2y^4)$$

$$@ f(x, y)$$

$$z = (x+y)(x-y) + 3$$

$$z = (x^2 - y^2) + 3$$

$$(b) (xy, 3x^2y^4)$$

$$z = (xy)(3x^2y^4) + 3$$

$$z = (3x^3y^5) + 3$$

$$g g(x) = x \sin x$$

$$(a) g(x/y)$$

$$z = xy \sin \frac{x}{y}$$

(b) $g(xy^2)$
 Solo-

$$= xy^2 \sin xy^2$$

(c) $g(x-y)$
 Solo-

13-1 $(x-y) \sin(x-y)$

Q5 :-

Find $F(g(x), h(y))$

if $F(xy) = xe^{xy}$, $g(x) = x^3$ and $h(y) = 2y - 1$

Q6) :-

Find $g(u(xy), v(xy))$

if $g(xy) = y \sin(x^2y)$, $u(xy) = x^3y^3$ and $v(xy) = \sqrt{xy}$

$$v(xy) = \sqrt{xy}$$

Q7): Let $F(x, y) = x + 3x^2y^2$, $x(t) = t^2$ and $y(t) = t^3$, Find

(a) $f(x(t), y(t))$

(b) $f(x(0), y(0))$

(c) $f(x(1), y(1))$

Q8) let $g(xy) = ye^{-3x}$

and $x(t) = \ln(t^2 + 1)$ and

Find $y(t) = t+1$
 Find $g(x(t)) \rightarrow y(t)$

Q5)

Sol:-

$$F(g(x), h(y)) = g(x)e^{g(x)h(y)}$$

$$= x^3 e^{x^3(2y-1)}$$

Q6)

Sol:-

$$g(u(xy), v(xy)) = v(xy) \sin u(xy)$$

$$= v(xy) \sin u(xy)^2 v(xy)$$

$$= \pi xy \sin(x^3 y^3)^2 (\pi xy)$$

$$= \pi xy \sin(x^6 y^6) (\pi xy)$$

$$= \pi xy \sin \pi xy$$

$$x + 3x^2y^0, x(0)^0$$

Q7) (a) $f(x(t), y(t))$
Sol:-

$$= x(t)^0 + 3(x(t))^2 (y(t))^0$$

$$= xt + (3xt^2)(y^0)$$

$$= xt + 3xt^2y^0 \Rightarrow t^2 3(0 \cdot t^4 \cdot t^5)$$

$$= t^2 + 3(y^{24})$$

Q8) (b) $f(x(0), y(0))$
Sol:-

$$= x(0) + 3(0)^2 x(0)^2 y(0)^0$$

$$= 0$$

(c) $f(x(1), y(1))$
Sol:-

$$= x(1) + 3(x(1))^2 (y(1))^2$$

$$= x + 3x^2y^2$$

$$= 1 + 3(1)^2(1)^2$$

$$= 1 + 3$$

$$= 4$$

Q8) $g(x, y) = ye^{-3x}$
Sol:-

$$= g(x, y(t), y(t))$$

$$\Rightarrow g(x, y) = ye^{-3x}$$

$$= y(t) e^{-3x(t)}$$

$$= y(t+1) e^{-3x(t+1)}$$

Use limit laws and continuity properties to evaluate the limit.

$$1.) \lim_{(x,y) \rightarrow (1,3)} (4xy^2 + x)$$

Solⁿ:-

$$\begin{aligned} &= 4(1)(3)^2 + 1 \\ &= 4(9) + 1 \\ &= 37 \end{aligned}$$

$$2.) \lim_{(x,y) \rightarrow (1/4, \pi)} xy^2 \sin xy$$

Solⁿ:-

$$\begin{aligned} &= \frac{1}{4} (\pi)^2 \sin \left(\frac{1}{4}\right) \pi \\ &= \frac{\pi^2 \sin \frac{\pi}{4}}{4} \end{aligned}$$

$$3.) \lim_{(x,y) \rightarrow (-1,2)} \frac{xy^3}{x+y}$$

$$\begin{aligned} &= \frac{(-1)(2)^3}{-1+2(2)} \\ &= \frac{-8}{3} \end{aligned}$$

$$(4) \lim_{(x,y) \rightarrow (1,3)} e^{2x-y}$$

Solⁿ:-

$$= e^{2(1)-3}$$

$$= e^{2-3}$$

$$= e^{-1}$$

$$= e^{-1}$$

$$(5) \lim_{(x,y) \rightarrow (1,0)} \ln(1+x^2y^3)$$

Solⁿ:-

$$= \ln(1+1^2(0)^3)$$

$$= \ln(1)$$

$$(6) \lim_{(x,y) \rightarrow (4,2)} x \sqrt[3]{y^3+2x}$$

Solⁿ:-

$$= 4 \sqrt[3]{(-2)^3+2(-4)}$$

$$= 4 \sqrt[3]{-8+8}$$

$$= 4 \sqrt[3]{0}$$

$$= 0$$

Determine whether the limit exists, if so find its value.

(15)
(x

3) Limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^4}{x^2 - y^2}$

so

Sol:-

$$= \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 - y^2)(x^2 + y^2)}{(x^2 - y^2)}$$

$$= \lim_{(x,y) \rightarrow (0,0)} = 0^2 + 0^2 = 0$$

(14) Limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 16y^4}{x^2 - 4y^2}$

Sol:-

$$= 1 \cdot \frac{(x^2 - 4y^2)(x^2 + 4y^2)}{(x^2 - 4y^2)}$$

$$= 1 \cdot \frac{(0)^2 + 4(0)^2}{0} = 0$$

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$$(15) \lim_{(x,y,z) \rightarrow (3,0,-4)} \frac{xz^2}{\sqrt{x^2+y^2+z^2}}$$

$$\text{Sol: } \therefore z = 1 \quad \text{or} \quad \frac{(3)(-4)^2}{\sqrt{(3)^2 + (0)^2 + (-4)^2}}$$

$$= \frac{48}{\sqrt{9+0+16}} = \frac{48}{5}$$

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